MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

RESISTOR, PULLUP, POWER, KEY CODE ADL

This specification is approved for use within the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic pullup resistor module.

* 1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

* 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of the specification and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Space and Naval Warfare Systems Command, ATTN: SPAWAR 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.065 failures per million operating hours for class I or 0.222 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be ADL.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/216-1 for class I and M28787/216-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

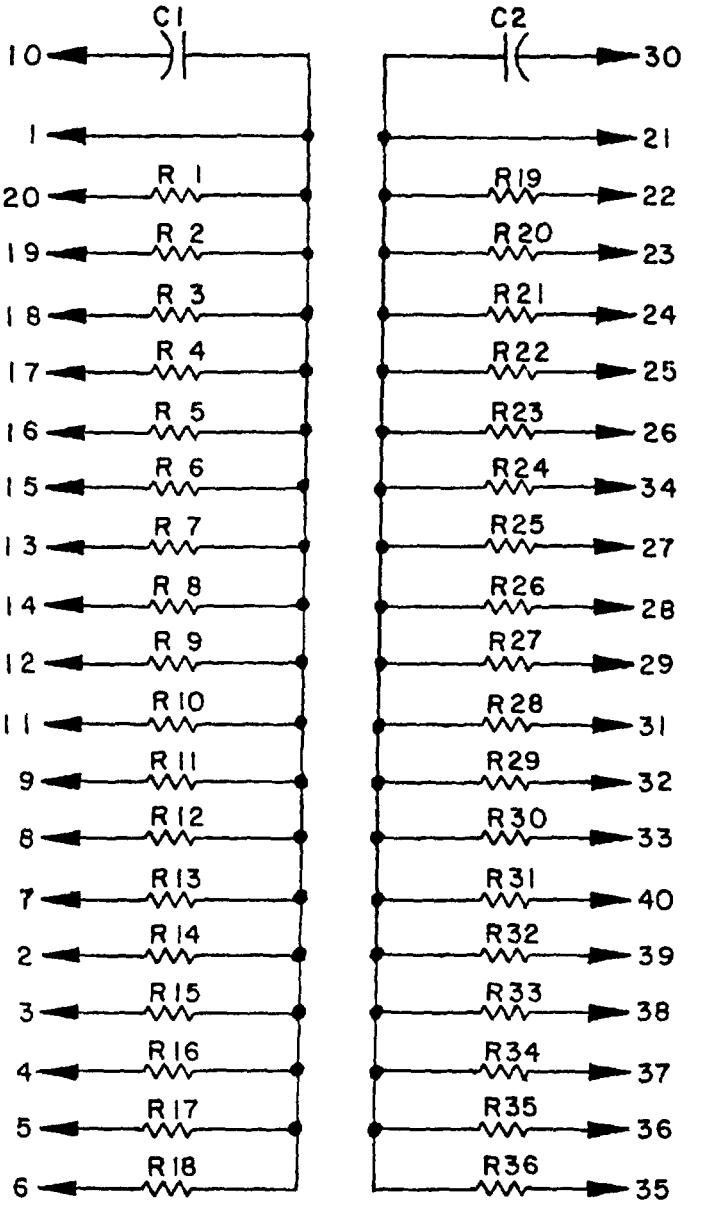
3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

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NOTES
1. R1 through R36 = 2,200Ω, 1/10 W.
2. C1 = C2 = 0.047 µF ±20%, 50 V dc.

* FIGURE 1 ADL module function diagram, pullup resistor

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★ TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1, A2	Isolation													
	Initial	II	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	II	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B36	Resistance													
	Initial	III	X	2,090	2,310	2,050	2,350	2,050	2,350	2,050	2,350	2,050	2,350	Ω
	End-of-life	III	X	2,035	2,365	2,024	2,376	2,024	2,376	2,024	2,376	2,024	2,376	Ω
C1, C2	Capacitance	IV	X	0.037	0.057	0.037	0.057	0.037	0.057	0.037	0.057	0.037	0.057	μF
C3, C4	Dissipation factor	IV	X	0	5	0	5	0	5	0	5	0	5	%
D1-D36	Crosstalk	V	Q	0	10	0	10	0	10	0	10	0	10	mV rms

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample accpetance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through D shall be performed in any order with the following exceptions. If test condition A is part of a sequence of tests, the order of testing shall be as follows: A where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through D in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B through D shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 2. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests B through D shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A through D shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

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6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

6.4 Cross reference of classification. Military specification MIL-M-28787/216 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2833694. Part number M28787/216-1 supersedes part numbers 2833694 and M28787/216.

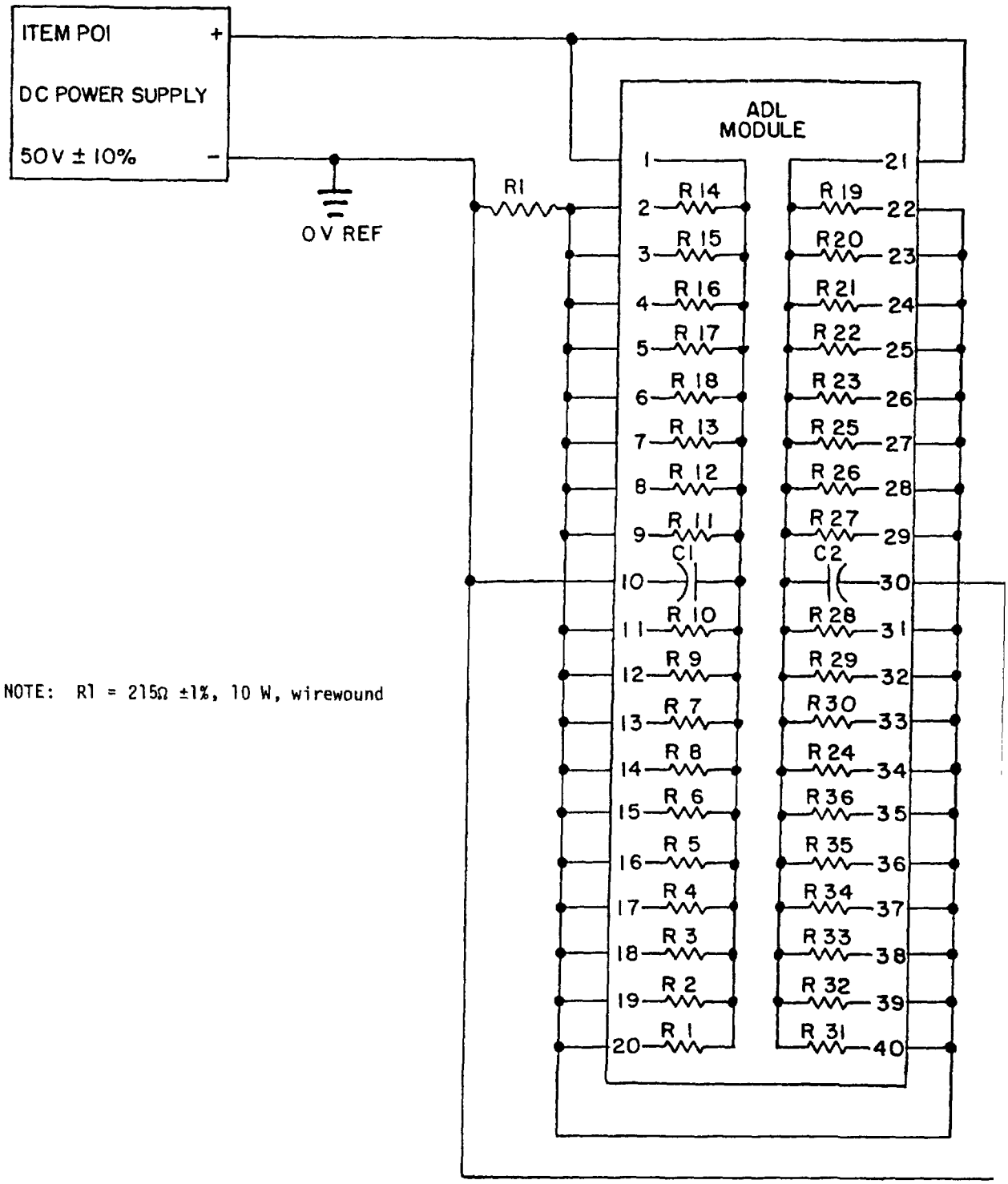
* 6.5 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

	Class I	Class II	Unit
Power dissipation	3.6 max	3.6 max	W
Resistor values			
Initial	2,200 ±5%	2,200 ±5%	Ω
End-of-life	2,200 ±5%	2,200 ±8%	Ω

* 6.6 Module function. This module contains 2 groups of 2,200 ohms, 1/10 watt resistors (18 resistors per group), each with a common terminal for connection to a power supply.

6.7 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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NOTE: R1 = 215Ω ±1%, 10 W, wirewound

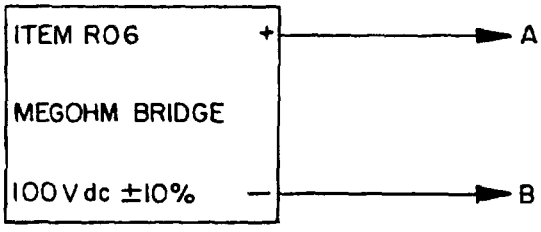
* FIGURE 2. Operating temperature, component temperature, vibration and life test circuit.

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TABLE II. Isolation tests.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Circuit to circuit	1 through 20	21 through 40	10	∞	MΩ	3
A2	All circuits to frame	1 through 40	Module frame	10	∞	MΩ	3

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

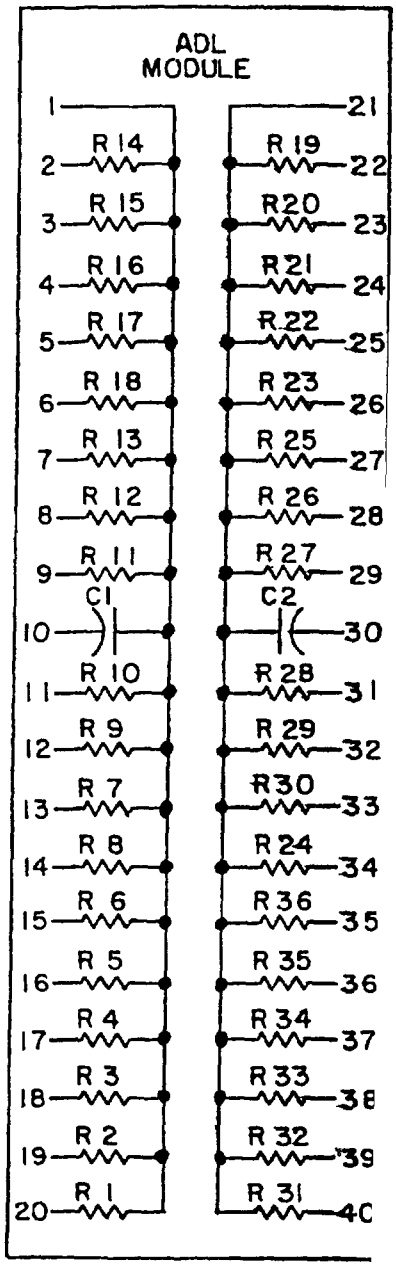


FIGURE 3. Isolation test circuit for table II.

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TABLE III. Resistance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	R14 Resistor	1	2	2,090	2,310	Ω	4
B2	R15 "	"	3	"	"	"	"
B3	R16 "	"	4	"	"	"	"
B4	R17 "	"	5	"	"	"	"
B5	R18 "	"	6	"	"	"	"
B6	R13 "	"	7	"	"	"	"
B7	R12 "	"	8	"	"	"	"
B8	R11 "	"	9	"	"	"	"
B9	R10 "	"	11	"	"	"	"
B10	R9 "	"	12	"	"	"	"
B11	R7 "	"	13	"	"	"	"
B12	R8 "	"	14	"	"	"	"
B13	R6 "	"	15	"	"	"	"
B14	R5 "	"	16	"	"	"	"
B15	R4 "	"	17	"	"	"	"
B16	R3 "	"	18	"	"	"	"
B17	R2 "	"	19	"	"	"	"
B18	R1 "	"	20	"	"	"	"
B19	R19 "	21	22	"	"	"	"

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TABLE III. Resistance test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
B20	R20	Resistor	21	23	2,090	2,310	Ω	4
B21	R21	"	"	24	"	"	"	"
B22	R22	"	"	25	"	"	"	"
B23	R23	"	"	26	"	"	"	"
B24	R25	"	"	27	"	"	"	"
B25	R26	"	"	28	"	"	"	"
B26	R27	"	"	29	"	"	"	"
B27	R28	"	"	31	"	"	"	"
B28	R29	"	"	32	"	"	"	"
B29	R30	"	"	33	"	"	"	"
B30	R24	"	"	34	"	"	"	"
B31	R36	"	"	35	"	"	"	"
B32	R35	"	"	36	"	"	"	"
B33	R34	"	"	37	"	"	"	"
B34	R33	"	"	38	"	"	"	"
B35	R32	"	"	39	"	"	"	"
B36	R31	"	"	40	"	"	"	"

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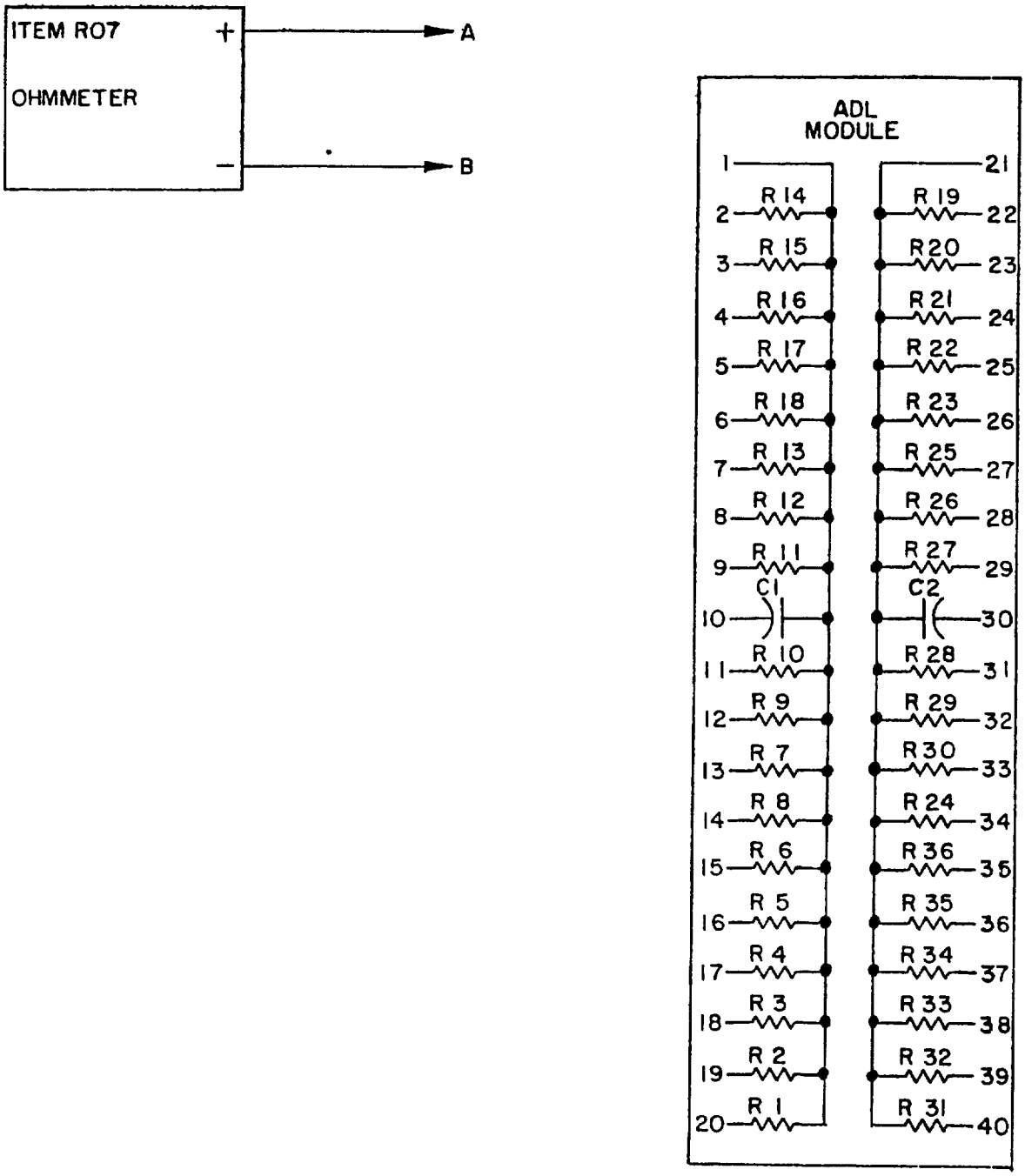


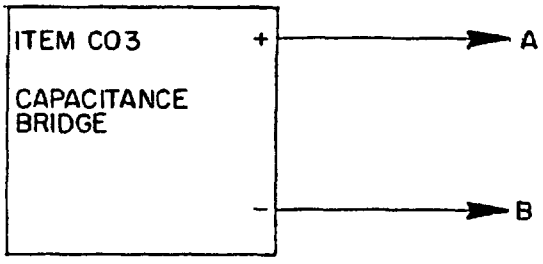
FIGURE 4. Resistance test circuit for table III.

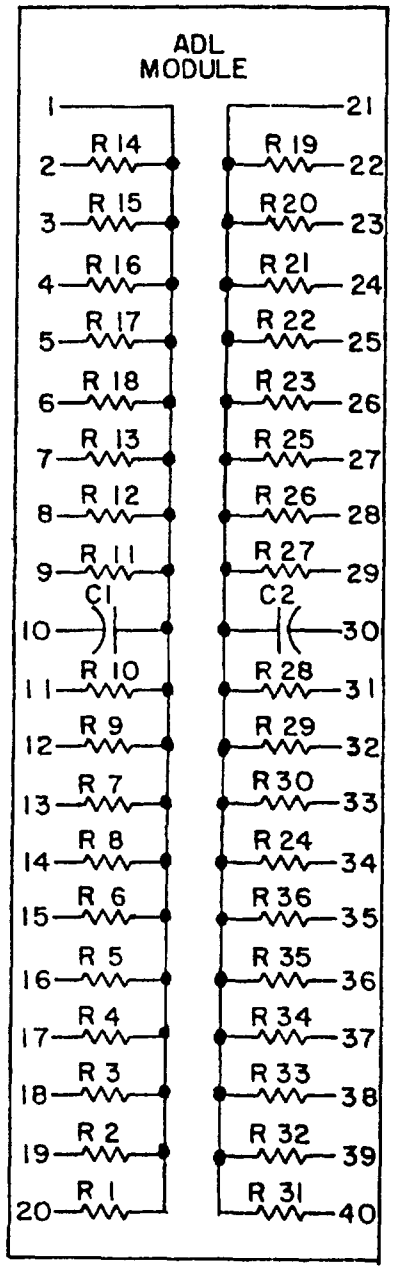
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TABLE IV. Capacitance and dissipation factor test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Capacitance C1	1	10	0.037	0.057	µF	5
C2	Capacitance C2	21	30	0.037	0.057	µF	"
C3	Dissipation factor C1	1	10	0	5	%	"
C4	Dissipation factor C2	21	30	0	5	%	"

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NOTE Item C03 setup
Frequency = 1 kHz ±100 Hz.
V = 1 V ±10%.

* FIGURE 5. Capacitance and dissipation factor test circuit for table IV.

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TABLE V. Crosstalk test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to all pins except	Min	Max		
D1	Crosstalk (circuit R14 to all other circuits)	2	1,2,10,21,30	0	10	mV rms	6
D2	Crosstalk (circuit R15 to all other circuits)	3	1,3,10,21,30	"	"	"	"
D3	Crosstalk (circuit R16 to all other circuits)	4	1,4,10,21,30	"	"	"	"
D4	Crosstalk (circuit R17 to all other circuits)	5	1,5,10,21,30	"	"	"	"
D5	Crosstalk (circuit R18 to all other circuits)	6	1,6,10,21,30	"	"	"	"
D6	Crosstalk (circuit R13 to all other circuits)	7	1,7,10,21,30	"	"	"	"
D7	Crosstalk (circuit R12 to all other circuits)	8	1,8,10,21,30	"	"	"	"
D8	Crosstalk (circuit R11 to all other circuits)	9	1,9,10,21,30	"	"	"	"
D9	Crosstalk (circuit R10 to all other circuits)	11	1,10,11,21,30	"	"	"	"
D10	Crosstalk (circuit R9 to all other circuits)	12	1,10,12,21,30	"	"	"	"
D11	Crosstalk (circuit R7 to all other circuits)	13	1,10,13,21,30	"	"	"	"
D12	Crosstalk (circuit R8 to all other circuits)	14	1,10,14,21,30	"	"	"	"
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TABLE V. Crosstalk test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to all pins except	Min	Max		
D13	Crosstalk (circuit R6 to all other circuits)	15	1,10,15,21,30	0	10	mV rms	6
D14	Crosstalk (circuit R5 to all other circuits)	16	1,10,16,21,30	"	"	"	"
D15	Crosstalk (circuit R4 to all other circuits)	17	1,10,17,21,30	"	"	"	"
D16	Crosstalk (circuit R3 to all other circuits)	18	1,10,18,21,30	"	"	"	"
D17	Crosstalk (circuit R2 to all other circuits)	19	1,10,19,21,30	"	"	"	"
D18	Crosstalk (circuit R1 to all other circuits)	20	1,10,20,21,30	"	"	"	"
D19	Crosstalk (circuit R19 to all other circuits)	22	1,10,21,22,30	"	"	"	"
D20	Crosstalk (circuit R20 to all other circuits)	23	1,10,21,23,30	"	"	"	"
D21	Crosstalk (circuit R21 to all other circuits)	24	1,10,21,24,30	"	"	"	"
D22	Crosstalk (circuit R22 to all other circuits)	25	1,10,21,25,30	"	"	"	"
D23	Crosstalk (circuit R23 to all other circuits)	26	1,10,21,26,30	"	"	"	"
D24	Crosstalk (circuit R25 to all other circuits)	27	1,10,21,27,30	"	"	"	"

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TABLE V. Crosstalk test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to all pins except	Min	Max		
D25	Crosstalk (circuit R26 to all other circuits)	28	1,10,21,28,30	0	10	mV rms	6
D26	Crosstalk (circuit R27 to all other circuits)	29	1,10,21,29,30	"	"	"	"
D27	Crosstalk (circuit R28 to all other circuits)	31	1,10,21,30,31	"	"	"	"
D28	Crosstalk (circuit R29 to all other circuits)	32	1,10,21,30,32	"	"	"	"
D29	Crosstalk (circuit R30 to all other circuits)	33	1,10,21,30,33	"	"	"	"
D30	Crosstalk (circuit R24 to all other circuits)	34	1,10,21,30,34	"	"	"	"
D31	Crosstalk (circuit R36 to all other circuits)	35	1,10,21,30,35	"	"	"	"
D32	Crosstalk (circuit R35 to all other circuits)	36	1,10,21,30,36	"	"	"	"
D33	Crosstalk (circuit R34 to all other circuits)	37	1,10,21,30,37	"	"	"	"
D34	Crosstalk (circuit R33 to all other circuits)	38	1,10,21,30,38	"	"	"	"
D35	Crosstalk (circuit R32 to all other circuits)	39	1,10,21,30,39	"	"	"	"
D36	Crosstalk (circuit R31 to all other circuits)	40	1,10,21,30,40	"	"	"	"

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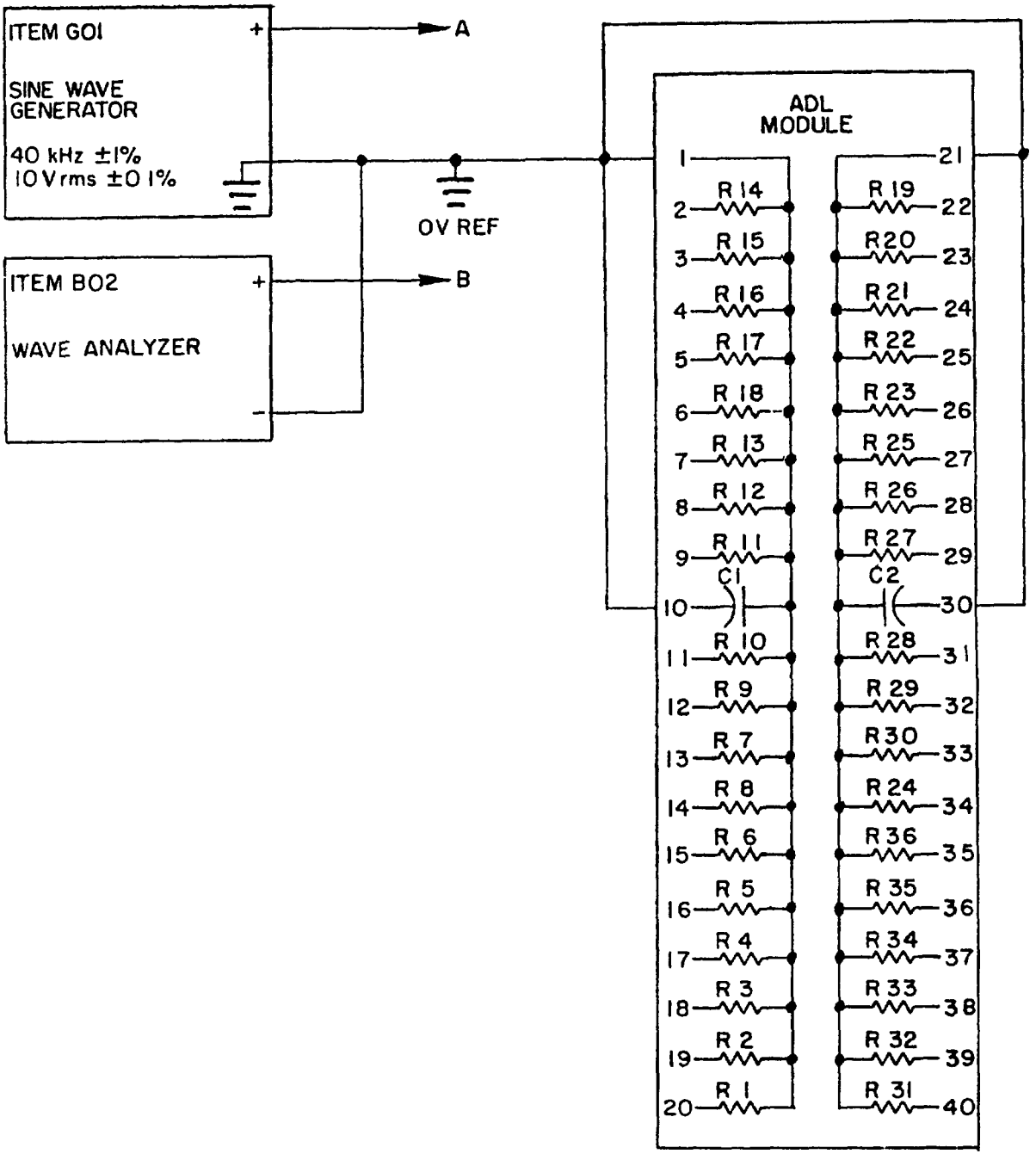


*FIGURE 6. Crosstalk test circuit for table V.

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